

TARK SIH26007

MASTER PROJECT DOSSIER

Complete System Software Hardware Electrical Communication Safety Validation and Deployment Reference

Research Prototype

Not Mine Certified

Human Operator Remains Responsible

Document control field	Controlled value
Project	TARK SIH26007
Problem statement	Safe and Efficient Operation of Mine Vehicles in Fog and Low Visibility Conditions in Open Cast Iron Ore Mines
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Category	Hardware
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Public deployment	TARK SIH26007 public simulation and monitoring demonstration
Current public URL	https://tark-sih26007-demo.onrender.com

Purpose. This dossier is the main human reference for the TARK SIH26007 research prototype. It identifies the controlled electrical design, software baseline, build boundaries, commissioning sequence, evidence status and supporting files. It does not authorize physical traction, field operation, certification or substitution of the physical emergency stop by software.

Deployment record. The project owner reported that the stated public URL was manually opened after the final Render deployment. This dossier records it as a public simulation and monitoring demonstration only. It is not evidence of physical vehicle operation or hardware verification.

Source Authority and Use

Authority order is: latest released electrical schematic for physical wiring; V8 Final Engineering Master Report for system architecture and safety philosophy; exact purchased component datasheet for real part limits and connector details; LEGO Style Build and Commissioning Manual V2 for physical sequence; and the frozen repository and closure reports for software behavior and test evidence. Where sources conflict, the higher authority controls and the lower source must be marked superseded or held for review.

Source type	Authority used in this dossier	Use
Controlled electrical presentation	FINAL_ENGINEERING_PRESENTATION.pdf and controlled SVG	Pin, terminal, net and physical E-stop reference
V8 master report	TARK SIH26007 Final Engineering Master Report V8	ODD, architecture, safety and validation intent
Build manual	TARK SIH26007 LEGO Style Build and Commissioning Manual V2	Assembly, labeling and power-off verification sequence
Frozen repository	commit 2d319e7 and tag tark-software-freeze-2026-09-20	Implemented software and test evidence
Purchased datasheets	Exact component variant documentation	Required before electrical or mechanical values are finalized

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1 Project Identity

TARK SIH26007 is a scaled research prototype for safe and efficient operation of mine vehicles in fog and low visibility conditions in open-cast iron ore mines. It is a safety-assistance engineering demonstration. It is not an autonomous mining vehicle and it is not a mine-certified safety system.

2 Problem Statement

Fog and low visibility reduce the reliability of a single visual sensor and make it difficult for an operator to judge whether the vehicle can continue safely. The engineering problem is therefore not only detecting an object. It is determining whether the perception available at that moment is sufficiently healthy, fresh and observable to support a bounded operating decision.

3 Project Objective

The objective is to combine complementary sensing, health, uncertainty, relative motion, collision risk and stopping margin so the system can restrict operation when the available perception is insufficient. Approved high-level safety states are NORMAL, WARN, RESTRICT, STOP and UNKNOWN where the evidence is insufficient. NORMAL does not mean certified safe; it means the current configured evidence supports the configured state.

4 System Concept

The contribution is system-level safety reasoning rather than a claim that a single sensor is novel. The controlled reasoning chain is: sensing -> health -> observability -> perception -> tracking -> uncertainty -> relative motion -> TTC -> stopping requirement -> PV SOE -> permitted speed -> safety state -> bounded command -> verification.

5 Overall Architecture

Primary information path: sensors -> Raspberry Pi 4 -> perception and tracking -> risk and PV SOE -> safety state -> bounded command -> ESP32-S3 -> Cytron MDD10A -> motors. Parallel physical path: physical normally closed E-stop -> contactor coil control -> K1 main traction contact -> traction isolation. The physical E-stop is independent of the Raspberry Pi and ESP32 software path.

Localization path: LC29H AA GNSS receiver -> GNSS driver -> normalized location service -> vehicle location and map context. GNSS is observational and map-context information. It is not the primary collision safety authority and never commands motors.

6 Operating Principle

The backend evaluates sensor age, health, detections, tracks and risk. It calculates a perception-verified envelope and issues only a bounded command. The ESP32 protocol supervisor validates the envelope, identity, version, sequence and expiry. In this frozen Phase 1 release the applied left and right output values remain zero and traction is DISABLED_PHASE_1.

7 Safety Philosophy

- No browser dashboard, public URL, map, GNSS input or sensor adapter has motion authority.
- A fault can preserve or reduce available capability; it must not silently increase it.
- The physical E-stop and K1 contactor path remain independent hardware protection.
- Simulation, replay, REAL transport reports, NOT_CONNECTED and FAULT are distinct source states. REAL software transport is not physical commissioning proof.
- All provisional PV SOE parameters remain PARAMETERIZED / NOT VALIDATED until a later controlled validation phase.

8 Phase Structure

Phase	Target	Traction and evidence status
Phase 1 50 percent	Complete software brain, simulator/replay and one radar software path	Traction disabled; software baseline frozen; physical evidence pending
Phase 2 70 percent	Vehicle hardware integration, drivers, encoders and sensor commissioning	Requires controlled component identity, datasheets and physical evidence
Phase 3 100 percent	System validation and demonstration under approved conditions	Not complete; no mine certification or field validation claimed

9 Complete Hardware Inventory

Section	Component	Qty	Purpose	Phase	Known status	Important verification
Computing	Raspberry Pi 4 Model B	1	Local backend, perception, dashboard service	Phase 1	SOFTWARE VERIFIED	Exact purchased revision and power input
Control	ESP32-S3 DevKitC-1	1	Protocol supervision and future interface boundary	Phase 1 and 2	SOFTWARE VERIFIED	Board identity, USB and flash
Radar	HLK-LD2450	1	Local obstacle/radar reports	Phase 1	SOFTWARE VERIFIED parser	Physical serial output and calibration
Thermal	MLX90640	1	Thermal frame boundary	Phase 2	SOFTWARE FUNCTIONAL	Breakout VCC and calibration
Camera	RGB UVC camera such as Logitech C920s	1	RGB frame boundary	Phase 2	SOFTWARE FUNCTIONAL	Exact model and UVC behavior
IMU	BNO055	1	Orientation and motion plausibility	Phase 2	SOFTWARE FUNCTIONAL	Breakout VCC and calibration
Localization	Waveshare LC29H AA and external active antenna	1 each	Vehicle location and map context	Phase 2	SOFTWARE FUNCTIONAL	Receiver identity and measured performance
Drive	JGB37 520 geared motors with Hall encoders	2	Differential drive and wheel response	Phase 2	HARDWARE PENDING	Exact encoder interface and pulse count
Motor driver	Cytron MDD10A Rev2.0	1	Dual motor interface	Phase 2	SOFTWARE FUNCTIONAL	Printed terminal markings and physical behavior
Power	12.8 V LiFePO4 battery with BMS	1	Prototype energy source	Phase 2	HARDWARE PENDING	Exact ratings and BMS documentation
Safety	Main protection, disconnect, NC E-stop, K1 contactor and driver	1 set	Physical traction isolation	Phase 2	HARDWARE PENDING	Fuse, coil, contactor and wiring ratings
Mechanical	4 wheel differential chassis, mounts, enclosure, connectors and test equipment	1 set	Prototype structure and assembly	Phase 2	HARDWARE PENDING	Exact purchased variant and dimensions

10 Computing System

The Raspberry Pi 4 runs FastAPI, sensor interfaces, perception, tracking, risk, event logging, replay, local WebSocket telemetry and the compiled React application. The ESP32-S3 is the bounded protocol and future actuation interface. The Pi and ESP32 communicate through the controlled Protocol V1 transport. Do not introduce an unreviewed Pi GPIO UART control path.

11 Perception System

The LD2450 target-report parser uses a bounded 30-byte target report with header AA FF 03 00, three eight-byte target records and footer 55 CC. It retains raw evidence, recovers from noise and rejects malformed reports and records outside the configured six metre envelope. X and Y are vehicle-local radar coordinates; they are never directly transformed into GNSS coordinates. Camera, thermal and IMU interfaces expose truthful health and are not permitted to fabricate samples.

12 Localization System

The target localization hardware is Waveshare LC29H AA with an external active antenna. The normalized contract separates GNSS, SIMULATION, REPLAY and UNKNOWN sources, and exposes NO_RECEIVER, SEARCHING, NO_FIX, 2D_FIX, 3D_FIX, STALE, ERROR and ONLINE states with reasons. No RTK fixed state, centimetre accuracy or physical GNSS performance is claimed.

13 Drive and Motion System

The intended prototype is a four-wheel differential-drive chassis with two JGB37 520 geared motors and Hall encoders. The current software uses the term wheel response, not ground-truth vehicle speed. Motor current, motor wiring polarity, encoder VCC, pulse count, polarity, counts per revolution and physical direction remain TBD / VERIFY AGAINST PURCHASED DATASHEET.

14 Motor Driver

Cytron MDD10A Rev2.0 is the controlled dual-motor driver target. The project has an MDD10A software interface and simulator, but no physical operation claim. Use the printed markings on the purchased board; do not infer terminal orientation from a diagram.

15 Power System

Traction and logic power are separate branches. Battery power passes through BMS and main protection. The traction positive path is gated by K1 main contact before MDD10A POWER+. Logic power passes through the protected logic branch and logic fuse to the 12.8 V to 5 V DC-DC converter, then to N-LOGIC5+. Exact fuse ratings, battery ratings, converter rating, wiring gauge, connector type and DC-DC topology remain TBD / VERIFY.

16 Emergency Stop and Traction Isolation

The control path is N-LOGIC12+ -> F3 -> ES1 normally closed physical E-stop -> CD1 contactor driver -> K1 coil -> driver return TBD / VERIFY. Coil suppression is connected across the two coil terminals. K1 physical actuator and K1 main contact have a black dashed mechanical linkage only; it is not an electrical connection. K1 auxiliary status through conditioning to ESP32 GPIO13 is diagnostic only and must never be described as the primary E-stop.

17 Electrical Architecture

Zone 1 provides battery, BMS, main fuse and protected 12.8 V branches. Zone 2 is a DC-DC converter with four independently represented terminals: IN+ from N-LOGIC12+, IN- from N-LOGIC12- / VERIFY, OUT+ to N-LOGIC5+, and OUT- to N-LOGIC-GND. Never assume converter input and output negatives are common. Junction dot means electrical connection only; black dashed line means physical or mechanical relationship only; wire crossings without a junction dot are not connected.

18 Raspberry Pi Pin to Pin Connections

Pi physical pin	Signal	Exact destination or use	Status
1	3V3	NO EXTERNAL POWER	Controlled restriction
2	5 V	N-LOGIC5+ regulated rail	VERIFY regulated rail and Pi power method
3	GPIO2 SDA	MLX90640 SDA and BNO055 SDA	VCC for sensor breakouts TBD / VERIFY
5	GPIO3 SCL	MLX90640 SCL and BNO055 SCL	VCC for sensor breakouts TBD / VERIFY
6	GND	N-LOGIC-GND	Controlled
8	GPIO14 TX	LD2450 Pin 4 RX	Controlled
10	GPIO15 RX	LD2450 Pin 3 TX	Controlled
20	GND	Logic ground reference	Controlled

LD2450: Pin 1 VCC -> regulated Pi 5 V logic rail; Pin 2 GND -> logic ground; Pin 3 TX -> Pi Pin 10 GPIO15 RX; Pin 4 RX -> Pi Pin 8 GPIO14 TX. UART is 256000 baud, 8 data bits, no parity, 1 stop bit. Use a voltage-compatible interface only after the purchased module documentation is checked.

19 ESP32 Pin to Pin Connections

ESP32 J1	GPIO or rail	Controlled destination	Meaning
4	GPIO4	Left encoder A	Encoder input
5	GPIO5	Left encoder B	Encoder input
6	GPIO6	Right encoder A	Encoder input
7	GPIO7	Right encoder B	Encoder input
15	GPIO9	MDD10A P4 PWM1	Left channel PWM mapping
16	GPIO10	MDD10A P5 DIR1	Left channel direction mapping
17	GPIO11	MDD10A P2 PWM2	Right channel PWM mapping
18	GPIO12	MDD10A P3 DIR2	Right channel direction mapping
19	GPIO13	K1 auxiliary diagnostic via approved conditioning if present	Diagnostic only
21	5 V	Logic rail connection	Controlled rail
22	GND	N-LOGIC-GND	Controlled rail

Unused ESP32 pins. J1 pins not listed in the controlled mapping are not allocated by this dossier. Do not reuse an unused pin because it is convenient. Any new allocation requires a revised controlled schematic and software review.

20 MDD10A Pin to Pin Connections

MDD10A point	Marking	Exact connection	Purpose
P1	GND	N-LOGIC-GND	Logic reference
P2	PWM2	ESP32 J1-17 GPIO11	Right channel PWM
P3	DIR2	ESP32 J1-18 GPIO12	Right channel direction
P4	PWM1	ESP32 J1-15 GPIO9	Left channel PWM
P5	DIR1	ESP32 J1-16 GPIO10	Left channel direction
T1	M1B	Left motor M1B	Left motor terminal
T2	M1A	Left motor M1A	Left motor terminal
T3	POWER+	K1 main contact output after E-stop gate	Traction positive
T4	POWER-	N-TRACTION- high-current return	Traction return
T5	M2A	Right motor M2A	Right motor terminal
T6	M2B	Right motor M2B	Right motor terminal

T3 is not connected before the physical E-stop path is verified. T4 remains the black high-current return, not blue signal wiring. Board orientation and connector polarity must be read from the purchased MDD10A marking and manufacturer documentation.

21 Sensor Wiring

Sensor	Controlled interface	Known connection	Hold point
LD2450	UART	Pi pins 8 and 10 as listed above; 256000 8N1	Module signal voltage and exact purchased connector
MLX90640	I2C	SDA Pi pin 3; SCL Pi pin 5	VCC, breakout voltage, pullups and address
BNO055	I2C	SDA Pi pin 3; SCL Pi pin 5	VCC, breakout voltage, address and calibration
RGB camera	USB	Raspberry Pi USB port	Exact UVC device and cable
LC29H AA	USB serial	Existing GNSS reader boundary	Verified serial path, baud and NMEA evidence
Encoders	ESP32 GPIO	Left A J1-4; Left B J1-5; Right A J1-6; Right B J1-7	VCC, interface, polarity and pulse count

22 Communication Architecture

GNSS path: reader -> checksum-validated normalized fix -> GnssLocationService -> vehicle-location API -> WebSocket -> frontend map. Radar path: adapter -> normalized report -> health and perception pipeline. Pi to ESP32 path: command -> framing -> COBS -> CRC32C -> canonical CBOR -> validation -> configured transport -> ACK/NACK/STATUS feedback. Camera, IMU and thermal follow adapter -> normalized data and health contract -> API/dashboard. These are separate paths; no duplicate authority path is permitted.

23 ESP32 Protocol

Protocol V1 is the single authoritative binary protocol. A wire frame is 00 | COBS encoded body and CRC | 00. The decoded body has magic 0x544B, version 1, type, payload length, sequence, timestamp, canonical CBOR payload and CRC-32C. Maximum payload is 512 bytes and maximum fully framed object is 640 bytes. The host and firmware share canonical test vectors for COMMAND, ACK, NACK, STATUS and HEARTBEAT.

Type	Required behavior	Safety meaning
COMMAND	Version, identity, sequence, timestamp, expiry, state, bounded values and configuration hash validated	Can be accepted only by software supervisor; Phase 1 applied values still zero
ACK	Accepted state, reason, output state, applied values and source mode	Does not prove physical motor output
NACK	Deterministic rejection reason and disabled output state	Malformed, expired or mismatched command is not repaired
STATUS	Telemetry-only reason and output state	Observation only
HEARTBEAT	Protocol version and configuration hash	Expiry or loss leads to safe disabled behavior

24 Software Architecture

The Raspberry Pi/backend contains configuration, typed contracts, hardware abstractions, sensor health, freshness, perception, tracking, uncertainty, TTC, stopping requirement, PV SOE, safety state, bounded command generation, Protocol V1 client, SQLite event persistence, replay, REST API and WebSocket. The React TypeScript frontend visualizes calculated data. The browser never performs PV SOE, drives motors or overrides constraints.

25 Perception Pipeline

Normalized radar reports are time-stamped, checked for freshness and passed into detection and tracking. Sensor health and uncertainty stay attached to the decision path. A malformed report is rejected and not converted into a detection. A sensor failure can lower confidence or produce UNKNOWN; it must not be presented as healthy live evidence.

26 Tracking

Tracks use stable identifiers, bounded history, age, position and relative motion where supported. Track association and lifecycle preserve uncertainty. Radar local position is maintained in a local vehicle frame and visualized separately from geographic map coordinates.

27 Uncertainty

Uncertainty represents limitations in sensing, freshness and interpretation. It is not decorative dashboard data. It constrains the available perception evidence and contributes to the permitted operating envelope. Exact field calibration and real sensor performance remain hardware validation work.

28 TTC and Risk

Time to collision is calculated from valid relative motion and is treated as UNKNOWN where inputs are invalid or insufficient. The dashboard displays backend-calculated TTC; it does not calculate or alter it. Negative, non-finite, stale or malformed data must not silently become valid risk evidence.

29 Stopping Requirement

Stopping requirement is a parameterized software calculation used with observable distance, reaction bound, deceleration bound and safety margin. The Phase 1 configuration records reaction bound 0.5 s, effective deceleration 0.5 m/s² and margin 0.5 m as PARAMETERIZED / NOT VALIDATED. These values are not physical brake-performance claims.

30 PV SOE

PV SOE means Perception Verified Safe Operating Envelope. It asks whether the current perception evidence supports a bounded operating condition. The computation produces envelope and stopping information, permitted speed, active constraints, reason and safety state. It is local backend logic; browser, map, GNSS and cloud paths cannot override it.

31 Safety State Machine

The approved states are NORMAL, WARN, RESTRICT, STOP and UNKNOWN. UNKNOWN is used where available evidence is insufficient. STOP and RESTRICT must be presented in text, shape and color; color alone is not sufficient. The Phase 1 hard speed cap is 0.0 m/s and traction remains DISABLED_PHASE_1.

32 Command Generation

The Pi produces a bounded command only after safety-state evaluation. The ESP32 supervisor validates sequence, expiry, heartbeat, version, configuration hash and payload. In Phase 1 accepted command output remains zero. There is no browser motor route, public API motor route or GNSS-to-motor path.

33 GNSS and Localization

The GNSS reader is a single bounded worker: serial bytes -> line framing -> checksum decoder -> RMC/GGA aggregation -> normalized service. It does not auto-select arbitrary devices. Non-simulation mode does not inject GnssSimulator data. Browser device location is intentionally separate and cannot populate vehicle-location state.

34 Map and Navigation

The MapLibre implementation uses OpenFreeMap with OpenStreetMap-compatible geographic data. It renders a vehicle marker, heading, accuracy radius, breadcrumbs, radar overlay, map controls and advisory route context where geometry exists. When vehicle location is unavailable it shows a labelled India overview. Map, reverse geocoding and routing provider failures must not interrupt the local safety loop.

35 Dashboard and HMI

The unified dashboard includes Dashboard, Map, Sensors, Safety, Logs, Replay, Settings and Diagnostics views. Driver, supervisor and owner/fleet role-aware views share one application. It is monitoring-only. It can show safety state, permitted speed, TTC, sensor health, communication health, simulation/replay status and evidence, but cannot command motors or bypass the E-stop.

36 Replay and Logging

Event persistence stores bounded records. Replay supports session catalog, selection, play, pause, step, seek, reset and end-of-replay. It distinguishes ORIGINAL RECORDING from REPLAY COMPUTATION and can report MATCH, MISMATCH or UNKNOWN. Replay output has no live actuation authority.

37 Simulation

Simulation is deterministic and labels generated sources SIMULATION. Supported scenarios include NORMAL, WARN, RESTRICT, UNKNOWN, STOP, radar loss, stale radar, communication loss, heartbeat loss, multi-target and target approach/recession. Simulation provides development evidence; it is not presented as live mine or vehicle data.

38 Fault Handling

Fault handling covers stale/malformed input, invalid timestamp/value/schema, protocol CRC and expiry errors, duplicate or out-of-order sequence, configuration mismatch, reconnect and unavailable devices. Bad data is rejected or marked fault; it must not silently become valid. All adapter lifecycle workers use bounded buffers and clean shutdown.

39 Security and Deployment

The deployed target serves the compiled frontend through FastAPI at one URL. Deployment configuration uses environment variables and .env.example rather than committed secrets. Map keys and provider credentials remain separated from source. The public demonstration is simulation and monitoring only; it does not expose motion authority.

40 Software Verification

Evidence	Frozen release result	Scope
Backend suite	97 passed	Backend, contracts, simulator, replay and integrations
Frontend suite	25 passed	Dashboard, state rendering, panels and replay
TypeScript	Passed	Frontend type correctness
Production build	Passed	Vite production build; MapView lazy-chunk advisory retained
Protocol V1 Python	10 passed	Vectors and transport interoperability
Firmware host tests	Passed	Protocol and service boundary; not board hardware

The results above are software verification. They do not verify a physical ESP32 flash, USB transport, MDD10A output, motor movement, encoder response, contactor, E-stop, sensor calibration, GNSS accuracy or vehicle behavior.

41 Public QA

Public QA commit 962365e fixed confirmed presentation defects: MapLibre default-style recovery after an external request blocker, responsive header behavior, unbroken NOT_CONNECTED and NORMAL_EVIDENCE tokens, camera unavailable spacing, event-list bottom padding, and removal of recording-creation controls from public demo mode. Navigation and browser console checks were reported clean for the scoped fix set.

42 Current Maturity Matrix

Area	Maturity	What is established	What is not established
Backend and safety pipeline	SOFTWARE VERIFIED	Core, PV SOE, logging, replay, API and simulation	Physical vehicle behavior
Protocol V1	SOFTWARE VERIFIED	COBS, CRC32C, canonical CBOR, vectors and host tests	Flashed board/USB/watchdog
LD2450 parser	SOFTWARE VERIFIED	Bounded target report decoder and simulated/runtime path	Physical device identity/calibration
GNSS	SOFTWARE FUNCTIONAL	Driver, parser, health, API and source gating	LC29H actual NMEA and accuracy
Camera IMU thermal	SOFTWARE FUNCTIONAL	Adapters, lifecycle, health and tests	Electrical compatibility/calibration
Drive and E-stop	HARDWARE PENDING	Pin contract and disabled software boundary	Powered motion, contactor and E-stop evidence
Public demo	OWNER REPORTED LIVE	Simulation monitoring URL supplied by owner	Independent external verification in this dossier build

43 Phase 1

Phase 1 is the software brain, simulation/replay, dashboard and radar software path with traction disabled. Its purpose is to prove the software architecture and evidence presentation without physical movement. The current hard cap is 0.0 m/s.

44 Phase 2

Phase 2 is controlled hardware integration: component identification, power-off wiring inspection, sensor bring-up, Pi and ESP32 discovery, encoder interface identification and non-traction diagnostics. No component may be energized beyond its verified documentation and the controlled schematic.

45 Phase 3

Phase 3 is planned validation with approved procedures, evidence records, calibrated parameters and project supervision. It is not completed. No mine field testing, mine certification or production deployment is claimed by this dossier.

46 Prototype Build Procedure

- Before each connection, identify the exact purchased part, revision, label and datasheet. Mark mismatches HOLD.

- Keep traction isolated and battery disconnected during assembly. Label both ends of every wire before attaching it.
- Build the physical power architecture first: battery and BMS, main protection, protected logic branch, DC-DC converter and logic rails. Verify with power off before energization.
- Install Raspberry Pi, ESP32, MDD10A, sensors and enclosure mounts without assuming orientation; use actual board markings.
- Complete controlled pin-to-pin connections in Sections 18 through 21. Meter-check continuity and isolation with power off.
- Install K1 E-stop control and traction path only after the correct contactor, driver, coil and protection datasheets are available.
- Record photos, part numbers, wire labels, inspection results and deviations in the build log.

47 Commissioning Procedure

- Start in simulation mode with traction disabled. Confirm backend, dashboard, health endpoint, WebSocket and event logging.
- Commission one hardware boundary at a time with all future traction wiring isolated. Begin with identity and raw diagnostic evidence.
- For GNSS, set only a confirmed device path; verify raw checksum-valid NMEA before accepting normalized fixes.
- For LD2450, capture raw serial evidence at 256000 8N1 before relying on normalized reports.
- For camera, IMU and thermal, verify the purchased device identity, documented supply/interface and a timestamped sample.
- For ESP32, verify board identity, flash and disabled Phase 1 status before connecting driver or motors.
- For encoders, verify exact electrical interface and counts with no traction; do not set encoder VCC until datasheet evidence exists.

48 Validation Procedure

Validation proceeds from software regression to bench evidence, subsystem evidence and controlled vehicle evidence. Each test requires a precondition, equipment, configuration hash, pass criteria, raw evidence, reviewer and outcome. A failed, unverified or contradictory result remains HOLD. Parameterized PV SOE values must be validated using measured conditions before any later traction authorization.

49 Test Records

Record	Current status	Evidence location
Frozen software suite	Passed	docs/SOFTWARE_FREEZE_BASELINE.md and docs/FINAL_SOFTWARE_COMMUNICATION_CLOSURE_REPORT.md
Public QA	Passed scoped QA	docs/PUBLIC_QA_FIX_REPORT.md
Hardware commissioning	NOT YET PHYSICALLY VERIFIED	docs/HARDWARE_ARRIVAL_CHECKLIST.md
Electrical wiring sign-off	HOLD until purchased components confirmed	FINAL_ENGINEERING_PRESENTATION.pdf and V2 build manual

50 Master BOM

The master bill of materials is the controlled hardware inventory in Section 9. It intentionally does not assign quantities, ratings, wire gauge, fuse rating, pricing or connector orientation where the exact purchased part is not frozen. Procurement must close these fields with manufacturer datasheets and project approval before a build step changes from HOLD.

51 Procurement Notes

Procurement records must capture exact manufacturer, model, revision, serial number where applicable, supplier, received date, datasheet link or stored copy, connector labeling and acceptance evidence. Any replacement component must be reviewed against the electrical schematic, software interface and mechanical mounting before installation.

52 Open Items

Open item	Required evidence	Current state
DC-DC input/output negative relationship	Exact converter datasheet and bench isolation check	TBD / VERIFY
Fuse values, wire gauge and connector ratings	Power calculation and purchased component documentation	TBD / VERIFY
K1 coil, contact rating and driver details	Exact contactor and driver datasheets	TBD / VERIFY
Encoder VCC, interface, pulses and polarity	Exact motor/encoder documentation and bench test	TBD / VERIFY
Sensor breakout voltage and address	Purchased MLX90640/BNO055 documentation	TBD / VERIFY
Mechanical dimensions and enclosure materials	Actual chassis and enclosure design record	TBD / VERIFY

53 Hardware Only Items

The following require physical evidence and are not software gaps: LC29H identity and accuracy; LD2450 physical serial output and calibration; ESP32 board identity, flashing and USB exchange; camera frame quality; BNO055 and MLX90640 calibration; encoder interface/counts; MDD10A and motor behavior; contactor/E-stop behavior; battery/BMS behavior; mounting; electromagnetic behavior; and vehicle-level validation.

54 Future Software Development

Future development may add verified hardware adapter settings, calibration workflows, field-data tools, performance work or later validated features. It must begin from a new branch or later commit based on the frozen tag. It must not overwrite or force-push the frozen baseline, and it must preserve the separation between monitoring, safety decision and physical E-stop authority.

55 File and Repository Index

The companion TARK_SIH26007_MASTER_PROJECT_INDEX.md lists the principal electrical, build, architecture, deployment, testing, source and firmware files with their workspace locations and role. It is an index, not an alternative authority hierarchy.

56 Change Control

Controlled electrical changes require a revised schematic and an explicit pin, terminal and net comparison. Software changes require a new branch or later commit, test evidence, review of safety boundaries and updated documentation. Changes to purchased components require datasheet reconciliation before wiring or configuration changes. Historical sources do not silently override the latest controlled source.

57 Final Project Status

Software and software communication are ready to freeze at commit 2d319e7 with tag tark-software-freeze-2026-09-20. The public demonstration is owner-reported as available at the recorded URL and is simulation and monitoring only. Hardware commissioning, electrical validation, component-specific ratings, physical E-stop evidence, motor behavior, sensor calibration, field validation and mine certification remain pending. TARK remains a research prototype.